Sai Srirama Saran Konala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.34
Sri Chaitanya Junior College	Class XII (APBIE)	2023	97.60%
Paramjyoti Public School	Class X (ICSE)	2021	94.00%

WORK EXPERIENCE

KLA Corporation - *AI/ML Intern*

- Developed custom **synthetic image generation** pipelines to address class imbalance in **SEM image datasets**, focusing on improving the defect detection model.
- Fine-tuned **Stable Diffusion** for class-specific SEM image generation, experimenting with custom losses, fine-tuning strategies, and pre/post-processing techniques.
- Developed a **3D-aware architecture** to synthesize consistent defect appearances across multiple viewing perspectives.
- Reduced **false positives by 27%** and improved **recall by 3%** in the downstream defect detection model.

ACHIEVEMENTS

- **Academic Excellence Award, IIT Hyderabad** - Awarded in recognition of exceptional academic performance.
- Achieved a perfect **10/10 SGPA** in Semester V.
- **5th Place**, Panchayati Raj Simulation - **Inter-IIT Tech Meet 12.0**.
- **6th Place**, IGDC Game Development - **Inter-IIT Tech Meet 13.0**.
- **Google Launchpad 2025 Mentee** - Was part of Google's student upskilling and mentorship program.
- **LPUNEST Scholar 2023** - Awarded a merit-based scholarship for academic excellence at the national level.

RESEARCH PROJECTS

Soccer Commentary Generation - *Dr. Manish Gupta (Microsoft Research) & Prof. Sandipan Dandapat (IITH)*

- Co-authored a **survey paper** analyzing the landscape of soccer commentary generation.
- Developed a **multimodal soccer commentary generation system** integrating Broadcast2Pitch GSR, T-DEED event spotting, YOLO Ball-Tracking, and GPT-5.4 to generate play-by-play commentary from broadcast video.

Diffusion Models for Language Modelling - *Prof. Saketha Nath (IITH)*

- Studied DDPM and foundational diffusion language models, including D3PM, Diffusion-LM, DiffuSeq, and MDLM.
- Implemented and compared 25M-parameter **autoregressive** and **masked diffusion language models (MDLM).**

PROJECTS

Face Recognition System for Automated Campus Security

- Developed a **one-shot** face recognition system using a **Siamese Neural Network**.
- Accelerated image retrieval by **50%** through **K-Means clustering** on facial embeddings for fast vector search.
- Integrated an anti-spoofing pipeline using image-to-depth mapping and facial landmark detection.

RISC-V Simulator

- Developed a **web-based RISC-V assembler and simulator**, implementing register, memory, and cache simulation, with error handling for smooth assembly-to-hex translation and execution.
- Built and deployed a **full-stack web interface** using React, NodeJS, and WebSockets enabling seamless real-time browser-based interaction with the simulator.

Inventory Management System

- Designed and implemented a **3NF MySQL database** with minimal redundancy and optimal performance.
- Developed efficient SQL **stored procedures**, **triggers**, and **user-defined functions** to handle inventory workflows.
- Built a **full-stack web application** using React and Express.js, with multi-user support and role-based access.

DeepStack

- Implemented **Transformer, DDPM, YOLOv1, DQN, and Policy Gradient** from scratch in PyTorch.
- Trained and evaluated models across **machine translation, generative modeling, object detection,** and **reinforcement learning** tasks.

SKILLS

Programming Languages: C/C++, Python, JavaScript, RISC-V Assembly

Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, ReactJS

Tools & Technologies: Git/GitHub, LaTeX, MySQL, Linux, Docker

RELEVANT COURSES

AI: Computer Vision, Natural Language Processing, Topics in DL, Deep Learning, Reinforcement Learning, Machine Learning

Computer Science: Data Structures, Algorithms, Computer Architecture, DBMS, Discrete Mathematics

Math: Convex Optimization, Probability and Random Variables, Linear Algebra, Statistics, Control Systems, Signals & Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Probability and Random Variables, IIT Hyderabad

Core Member, Epoch - AI & ML Club, IIT Hyderabad

Core Member, Lambda - Development Club, IIT Hyderabad

Machine Learning Member, Tinkerers' Lab, IIT Hyderabad

Tech Coordinator, Office of Career Services (OCS) - Placement Cell, IIT Hyderabad